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PyCCEA: A Python package of cooperative co-evolutionary algorithms for feature selection in high-dimensional data

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Summary

Feature selection is a critical preprocessing step in many machine learning pipelines, particularly when dealing with high-dimensional datasets commonly found in domains such as genomics, text mining, and image analysis. However, many feature selection techniques, such as heuristic search methods and evolutionary algorithms, struggle to maintain predictive performance and interpretability as the dimensionality of data increases (Theng & Bhoyar, 2024).

Cooperative co-evolutionary algorithms (CCEAs) offer a promising approach for tackling this challenge by dividing the high-dimensional space into multiple tractable low-dimensional subcomponents. Each subcomponent is evolved independently using a subpopulation, and candidate solutions are evaluated based on their collaboration with representatives from other subpopulations (Ma et al., 2018).

`PyCCEA` implements CCEAs specifically for feature selection in high-dimensional data. It provides modular components for decomposition, collaboration, fitness evaluation, and optimization, allowing users to reproduce existing methods or develop new ones. The framework currently only follows a wrapper-based approach, in which subsets of features are evaluated using machine learning models to guide the search. However, this design is not restrictive, `PyCCEA` can be extended to support other evaluation paradigms, such as filters, hybrid, or embedded methods, which are planned as future features.

All machine learning and data processing components are built on top of well-established libraries such as `scikit-learn` (Pedregosa et al., 2011) and `Pandas` (McKinney, 2010), ensuring compatibility, flexibility, and ease of use. To the best of our knowledge, `PyCCEA` is the first open-source package dedicated to CCEAs for feature selection, offering a standardized and extensible framework for tackling high-dimensional datasets in a reproducible manner, as shown in Figure 1.

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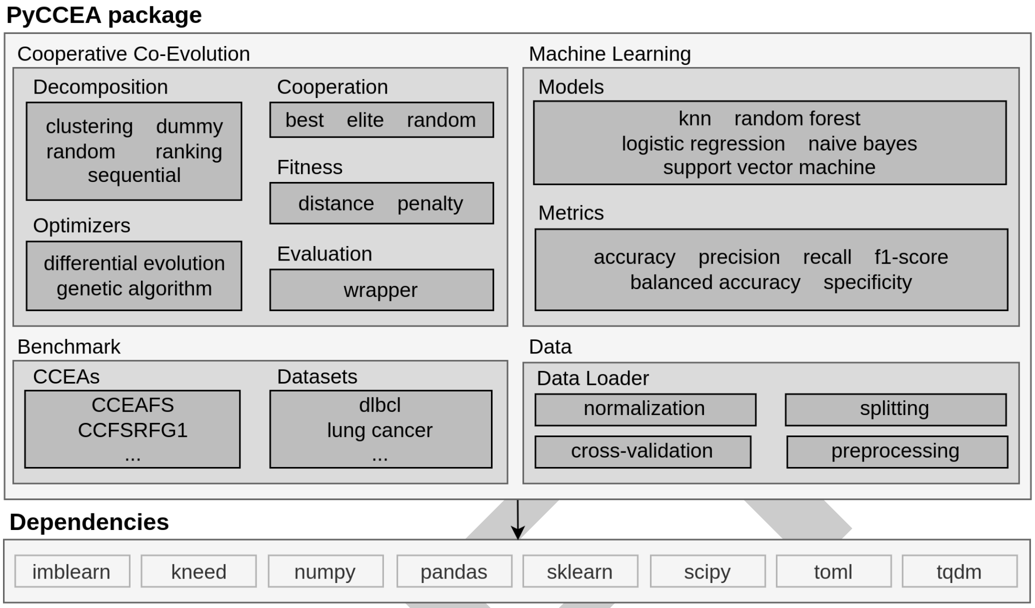

Figure 1: An overview of the PyCCEA package, its modules and underlying Python dependencies.

Statement of need

Despite the growing interest in cooperative co-evolutionary algorithms for feature selection and their promising results (Song et al., 2020), there is currently no publicly available software package that consolidates these techniques into a reusable and extensible tool. Existing research typically relies on custom (Firouznia et al., 2023; A. Rashid et al., 2020; A. B. Rashid et al., 2020) or unpublished implementations (Song et al., 2020; Zhou et al., 2024), which makes it difficult to reproduce results, compare methods, or build upon previous work.

PyCCEA addresses this gap by providing a well-organized, research-focused implementation of cooperative co-evolutionary algorithms tailored to feature selection. It incorporates widely used strategies from the literature and encourages standardization in experimental design and evaluation. By enabling consistent benchmarking and facilitating the development of new strategies, PyCCEA supports both methodological innovation and practical application in high-dimensional machine learning problems. Its release lowers the barrier to entry for researchers and practitioners, accelerating progress in the field of feature selection using evolutionary computation.

Recent research in evolutionary computation and feature selection has already leveraged PyCCEA as a foundational framework for implementing and evaluating cooperative co-evolutionary strategies (Venâncio & Batista, 2025), demonstrating its value in supporting reproducible and extensible scientific contributions. Future improvements aim to broaden its evaluation capabilities and encourage further community-driven development.

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